

MENTORA ACADEMY

DEPARTMENT OF ADVANCED CHEMISTRY · CAMBRIDGE INTERNATIONAL A LEVEL (9701)

Analytical Chemistry Experimental Suite

Planning, Analysis and Evaluation · TLC Separation of Analgesics · Spectroscopic Structure Deduction

CANDIDATE NAME:	Urwah	CENTRE NUMBER:	PK082	CANDIDATE NO:	0142
COMPONENT:	Paper 5 — Planning, Analysis and Evaluation	DURATION:	1 Hour 15 Minutes	MAX MARKS:	30 Marks

READ THESE INSTRUCTIONS FIRST

- Write your name, Centre number, and candidate number in the boxes above.
- Answer all questions in the spaces provided on the question paper.
- In Question 1 (Planning): define independent, dependent, and controlled variables; describe step-by-step procedures with apparatus diagrams; calculate required quantities; state hazard risk assessments.
- In Question 2 (Analysis, Conclusions & Evaluation): process raw experimental data into derived columns; quote values to consistent significant figures; plot graphs with appropriate linear scales; determine gradients and intercepts; identify anomalous points; evaluate uncertainties.
- You may use an approved scientific calculator. The Periodic Table is provided as standard reference.

EXAMINATION BREAKDOWN & SYLLABUS SKILLS TESTED

Question	Investigation Title & Focus	Syllabus Ref	Type of Task	Marks
Question 1	Planning a Thin-Layer Chromatography (TLC) Investigation to Identify Analgesic Drugs	9701/52/M/J/23/ Q1 · Syllabus 37.1	PLANNING	15
Question 2	Analysis and Evaluation of Multi-Spectroscopic Data to Deduce an Unknown Structure	9701/51/O/N/23/Q 2 · Syllabus 37.3 / 37.4	ANALYSIS & EVALUATION	15

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QUESTION 1

PLANNING

[Total: 15 Marks]

Planning a Thin-Layer Chromatography (TLC) Investigation to Identify Analgesic Drugs**Cambridge Syllabus Reference: 9701/52/M/J/23/Q1 · Syllabus 37.1**

An over-the-counter pain relief tablet is suspected to contain a mixture of paracetamol (4-acetamidophenol), aspirin (2-ethanoyloxybenzoic acid), and caffeine.

You are required to plan a thin-layer chromatography (TLC) investigation to separate and identify which of these active pharmaceutical ingredients are present in the crushed tablet.

You are provided with: pure reference samples of paracetamol, aspirin, and caffeine; a crushed tablet sample; silica gel TLC plates; an organic mobile phase (ethyl ethanoate / cyclohexane / ethanoic acid mixture); capillary micropipettes; a developing chromatography tank with lid; filter paper; and a UV viewing cabinet (254 nm).

- (a)** Describe how the crushed tablet sample is prepared into a concentrated liquid solution suitable for spotting onto a TLC plate. **[2]**

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- (b)** Detail the preparation of the TLC plate, including drawing the baseline, applying the spots, and labelling. **[3]**

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- (c)** Describe the development of the TLC plate in the chromatography tank, stating two essential precautions to ensure uniform solvent flow. **[3]**

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- (d)** Explain how the developed TLC plate is visualized and how the active ingredients in the tablet are identified. **[4]**

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- (e)** State what problem would occur if the mobile phase solvent level in the tank was higher than the pencil baseline. **[3]**

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QUESTION 2

ANALYSIS & EVALUATION

[Total: 15 Marks]

Analysis and Evaluation of Multi-Spectroscopic Data to Deduce an Unknown Structure

Cambridge Syllabus Reference: 9701/51/O/N/23/Q2 · Syllabus 37.3 / 37.4

An unknown neutral organic compound Z has the molecular formula $\text{C}_4\text{H}_8\text{O}_2$.

Elemental analysis confirms: C = 54.5%, H = 9.1%, O = 36.4%.

Infrared spectrum shows a strong, sharp absorption peak at 1740 cm^{-1} , but NO broad absorption between $2500\text{--}3300\text{ cm}^{-1}$ or $3200\text{--}3600\text{ cm}^{-1}$.

^{13}C NMR spectrum exhibits exactly 4 peaks at 14.1, 20.7, 60.4, and 171.2 ppm.

^1H NMR spectrum exhibits three distinct signals:

Signal 1: Triplet at 1.25 ppm (integration 3H)

Signal 2: Singlet at 2.05 ppm (integration 3H)

Signal 3: Quartet at 4.12 ppm (integration 2H)

- (a) Use the infrared data to identify the functional group present and state which functional groups are absent. [3]
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- (b) Analyze the ^{13}C NMR spectrum: explain why the peak at 171.2 ppm confirms an ester carbonyl, and state what the four peaks indicate about symmetry. [3]
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- (c) Analyze the ^1H NMR spectrum in detail. Explain the splitting pattern and chemical shift of Signal 1, Signal 2, and Signal 3 using the (n+1) rule. [5]
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- (d) Deduce the structural formula and systematic IUPAC name of Compound Z. [2]
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- (e) Predict how the ^1H NMR spectrum of methyl propanoate, $\text{CH}_3\text{CH}_2\text{COOCH}_3$, would differ from that of Compound Z. [2]
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Comprehensive Examiner Mark Scheme

Marking points, indicative responses, and error-carried-forward (ECF) guidelines

MARK SCHEME: QUESTION 1 (PLANNING) — Planning a Thin-Layer Chromatography (TLC) Investigation to Identify Analgesic Drugs

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Crush the tablet, dissolve in a small volume (2–3 cm ³) of ethanol / ethyl ethanoate to extract active ingredients [1]; Filter or centrifuge to remove insoluble tablet binders/fillers (e.g. starch) [1].	[2]
(b)	Draw a pencil baseline 1.5 cm from the bottom of the plate (never use ink which dissolves in mobile phase) [1]; Using fine glass capillary tubes, spot small concentrated droplets of tablet extract and each pure reference standard (aspirin, paracetamol, caffeine) along the baseline with at least 1 cm spacing [1]; Allow spots to dry completely before development [1].	[3]
(c)	Pour mobile phase into tank to a depth of ~0.5 cm (must be strictly below the pencil baseline) [1]; Precaution 1: Line tank walls with filter paper soaked in solvent and cover with tight lid to saturate atmosphere with solvent vapor (prevents uneven evaporation) [1]; Precaution 2: Do not move or disturb tank while solvent front ascends [1].	[3]
(d)	Remove plate when solvent front is ~1 cm from top, immediately mark solvent front with pencil, and dry plate [1]; Visualize colorless spots under UV lamp (254 nm) which reveals dark quenching spots against fluorescent background, and circle spots in pencil [1]; Measure distance from baseline to center of each spot and to solvent front, calculate $R_f = d_{\text{spot}} / d_{\text{front}}$ [1]; Match the R_f values and vertical positions of spots from the tablet lane with the pure reference standard lanes [1].	[4]
(e)	The sample spots would dissolve directly into the solvent pool at the bottom of the tank [2]; No separation would occur as solute would diffuse throughout the bulk solvent rather than ascending the plate [1].	[3]

MARK SCHEME: QUESTION 2 (ANALYSIS & EVALUATION) — Analysis and Evaluation of Multi-Spectroscopic Data to Deduce an Unknown Structure

[15 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Sharp peak at 1740 cm ⁻¹ indicates a carbonyl group (C=O), specifically an ester or aldehyde/ketone [1]; Absence of broad absorption at 3200–3600 cm ⁻¹ rules out alcohol (–OH) [1]; Absence of very broad absorption at 2500–3300 cm ⁻¹ rules out carboxylic acid (–COOH) [1].	[3]
(b)	171.2 ppm is characteristic of an ester carbonyl carbon (–COO–) deshielded by two electronegative oxygens [1]; Four distinct peaks indicate exactly 4 chemically non-equivalent carbon environments [1]; Since the formula is C ₄ H ₈ O ₂ , all four carbon atoms in the molecule have unique environments (no symmetry) [1].	[3]
(c)	Signal 1 (triplet at 1.25, 3H): Methyl group adjacent to 2 protons (–CH ₂ –): $n=2$, $n+1=3$ (triplet) [1]; Signal 3 (quartet at 4.12, 2H): Methylene group adjacent to 3 protons (–CH ₃): $n=3$, $n+1=4$ (quartet) [1]; Coupling between Signal 1 and 3 confirms an ethyl group (–CH ₂ CH ₃) [1]; Chemical shift of quartet at 4.12 ppm confirms –CH ₂ – is directly bonded to electronegative oxygen (–O–CH ₂ CH ₃) [1]; Signal 2 (singlet at 2.05, 3H): Isolated methyl group with no adjacent protons ($n=0$, singlet), bonded to carbonyl (CH ₃ –C=O) [1].	[5]
(d)	Structural formula: CH ₃ COOCH ₂ CH ₃ [1]; IUPAC name: Ethyl ethanoate [1].	[2]
(e)	In methyl propanoate, the singlet (3H) appears downfield at ~3.7 ppm (bonded to oxygen, –OCH ₃) [1]; The quartet (2H) appears upfield at ~2.3 ppm (bonded to carbonyl, –CH ₂ –C=O), opposite to ethyl ethanoate [1].	[2]